

Mechanical
Mechanical.SchDoc

Revision_History
Revision_History-1.SchDoc

Power_Supply
Power_Supply.SchDoc

VIRGINIA
INDUSTRY SOLUTIONS

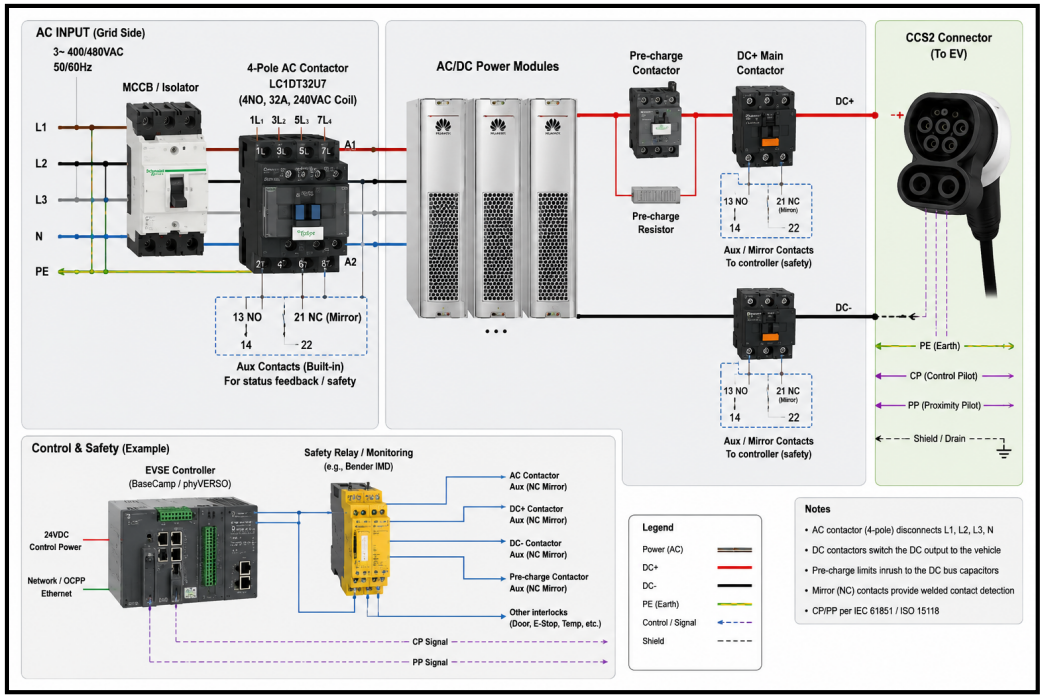
Title	***cb-vis-acdc-v1.0***		*Virginia industry solutions* *11166 Fairfax Blvd* *Fairfax* *USA, VA* *22030*
Size: A3	Number:1.	Revision:V1.0	
Date: 10/17/2026 Time: 11:05:31 AM	Sheet 1 of 13		
File: Hardware_Architecture.SchDoc			

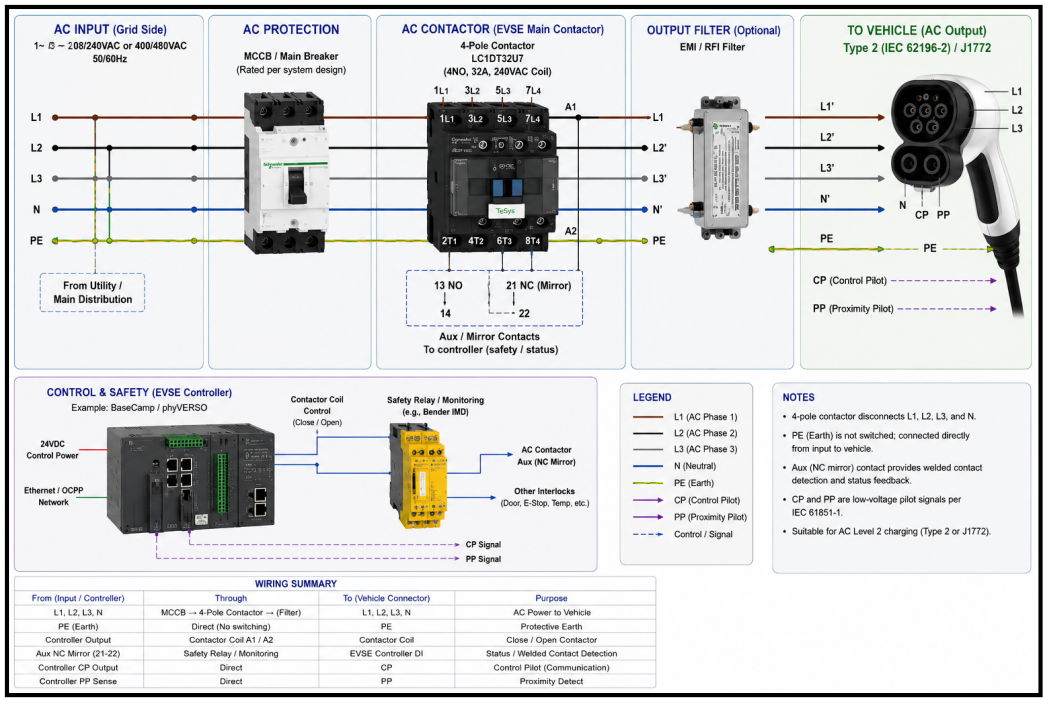
REVISION HISTORY

Revision V1.0

Initial release

Revision V2.0
CCU (Charge controller unit) seperated from the ACDC module





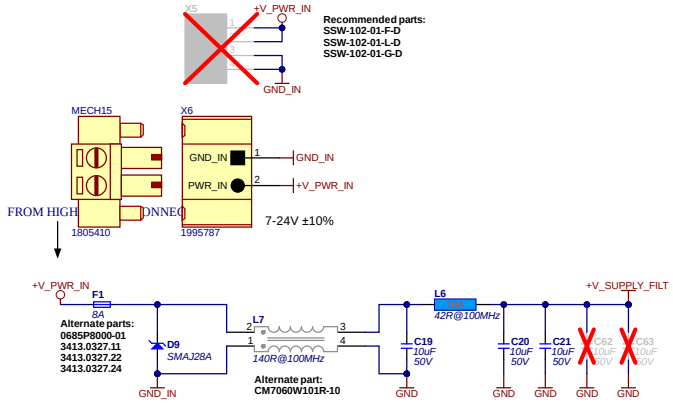
WIRING SUMMARY

From (Input / Controller)	Through	To (Vehicle Connector)	Purpose
L1, L2, L3, N	MCCB → 4-Pole Contactor → (Filter)	L1, L2, L3, N	AC Power to Vehicle
PE (Earth)	Direct (No switching)	PE	Protective Earth
Controller Output	Contactor Coil A1 / A2	Contactor Coil	Close / Open Contactor
Aux NC Mirror (21-22)	Safety Relay / Monitoring	EVSE Controller DI	Status / Welded Contact Detection
Controller CP Output	Direct	CP	Control Pilot (Communication)
Controller PP Sense	Direct	PP	Proximity Detect

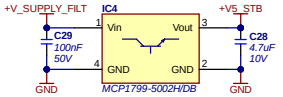
VIRGINIA INDUSTRY SOLUTIONS

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Size: A3	Number:2.	Revision:V1.0	*11166 Fairfax Blvd*
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File: Revision_History-1.SchDoc			*USA, VA*
			22030

POWER IN

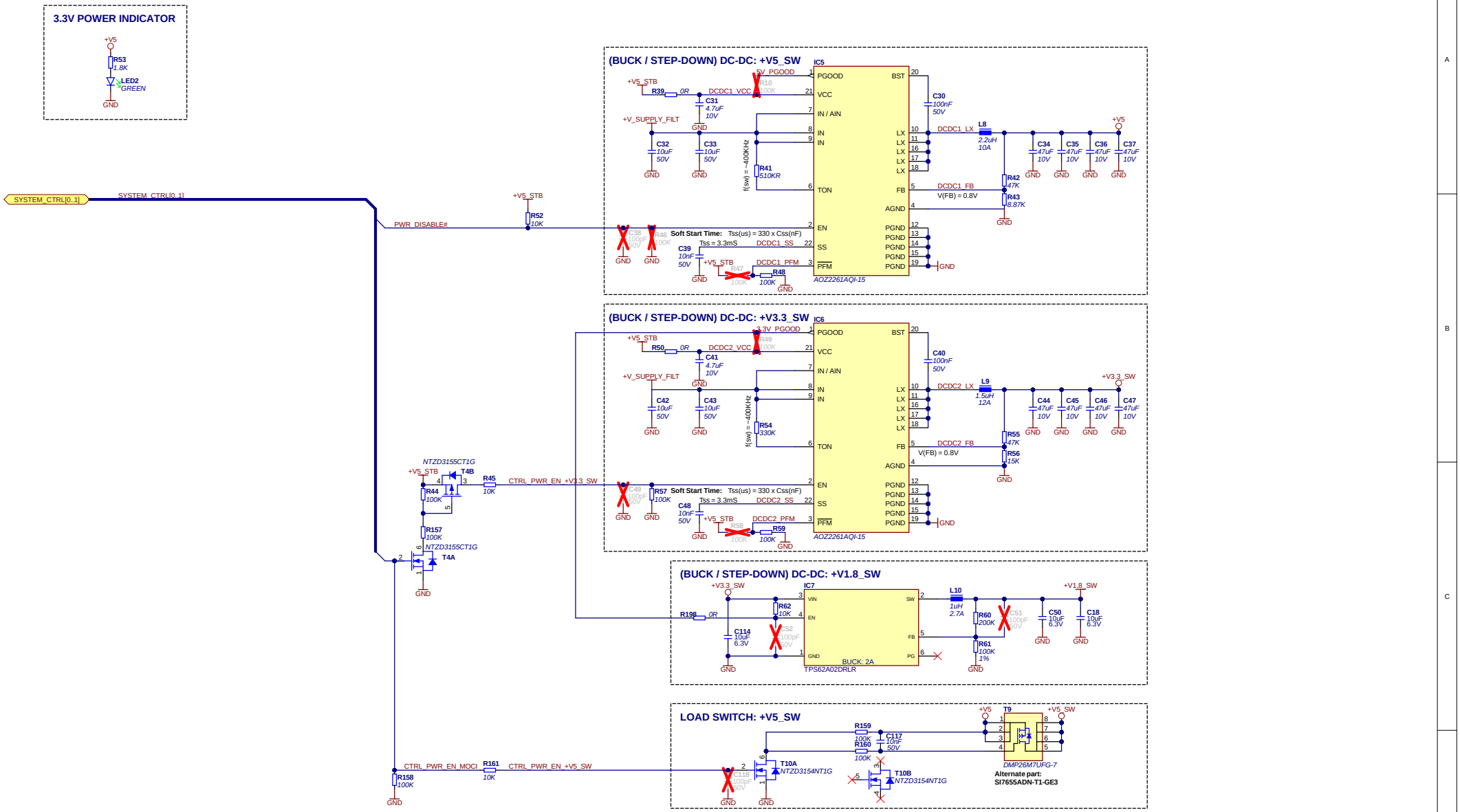


STANDBY VOLTAGE LDO

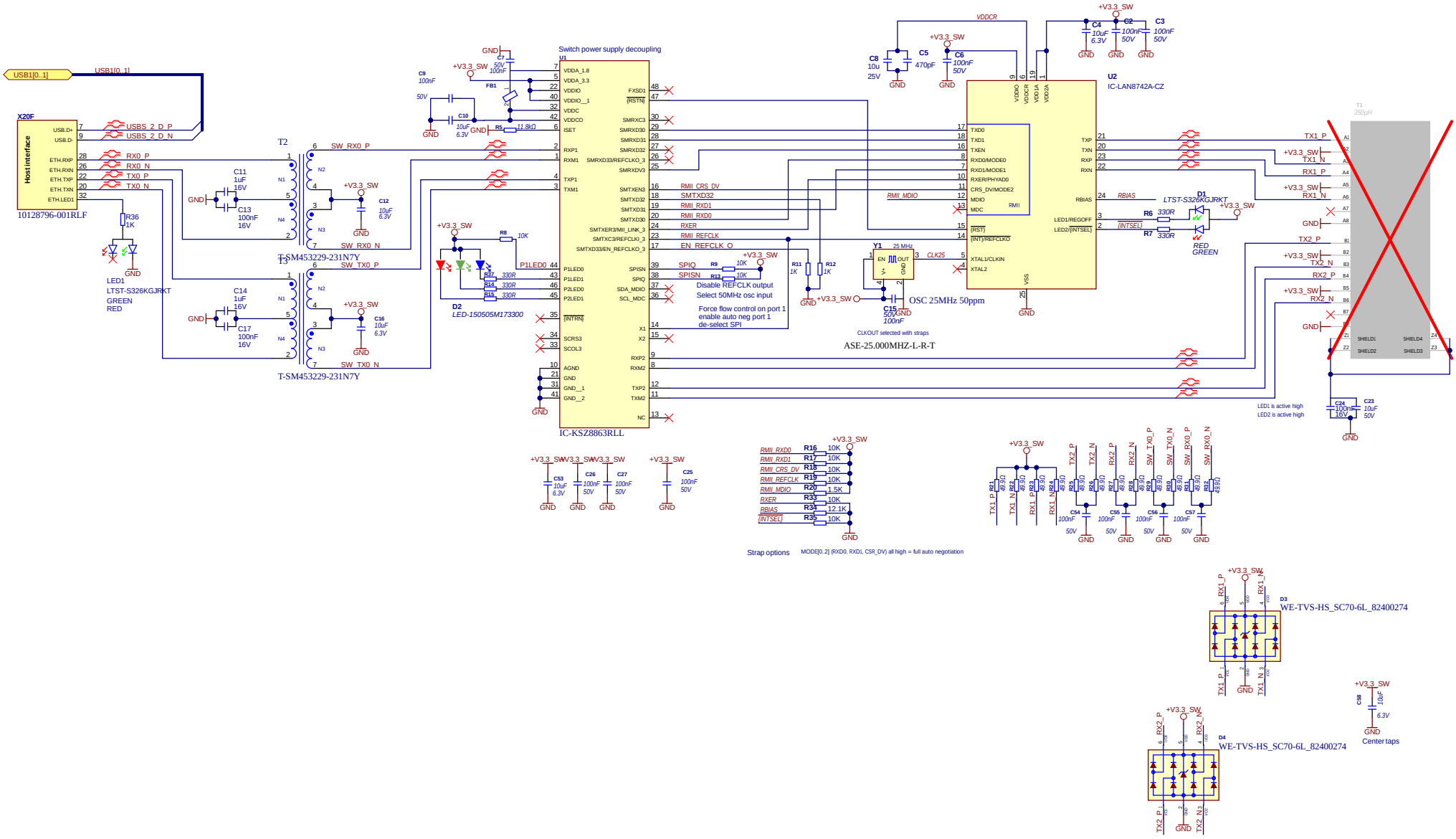


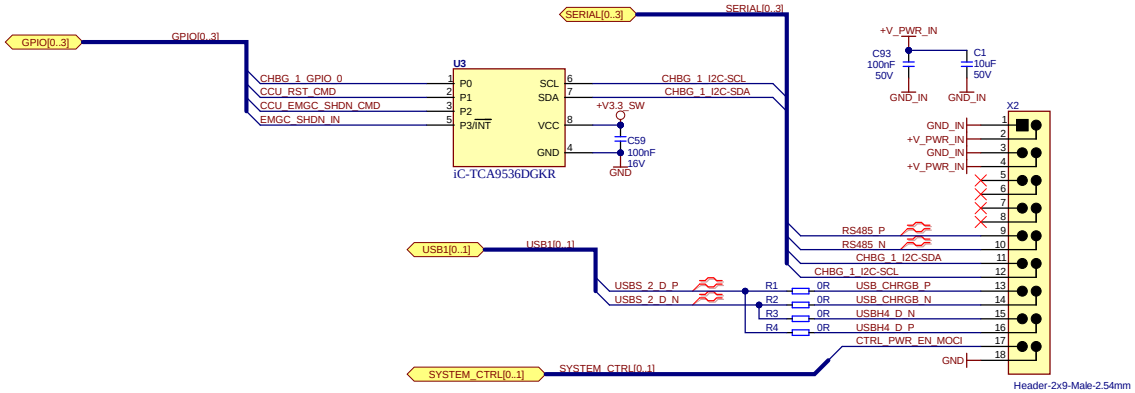
VIRGINIA INDUSTRY SOLUTIONS

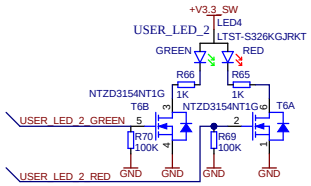
Title	cb-vis-acdc-v1.0		Virginia industry solutions 11166 Fairfax Blvd Fairfax USA, VA 22030
Size: A3	Number:3.	Revision:V1.0	
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File: Power_Supply.SchDoc			



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Size: A3	Number:4.	Revision:V1.0	
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File: Power_Switch.SchDoc			

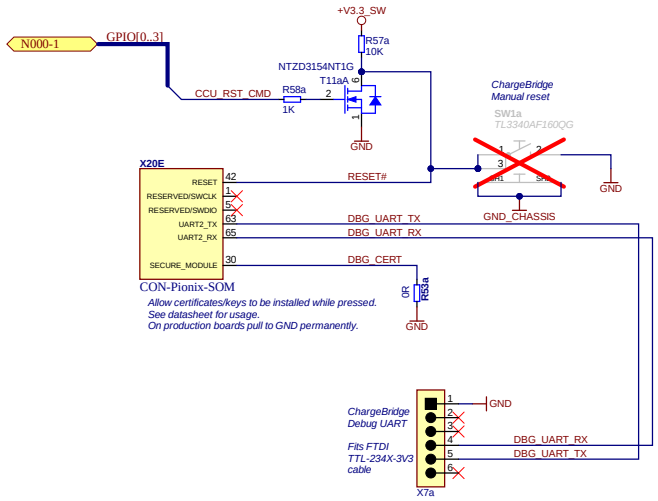

USB1[0..1]
X20F
Host interface
10128796-001RLF
T2
T-SM453229-231N7Y
T3
T-SM453229-231N7Y
LED1
LTST-S326KGJRKT
GREEN
RED
D2
LED-150505M173300
Switch power supply decoupling
U1
IC-KSZ8863RLL
U2
IC-LAN8742A-CZ
Y1
OSC 25MHz 50ppm
ASE-25.000MHZ-L-R-T
D1
LTST-S326KGJRKT
RED
GREEN
Disable REFCLK output
Select 50MHz osc input
Force flow control on port 1
enable auto neg port 1
de-select SPI
CLKOUT selected with straps
LED1 is active high
LED2 is active high
Strap options
MODE[0..2] (RXD0, RXD1, CSR_DV) all high = full auto negotiation
D3
WE-TVS-HS_SC70-6L_82400274
D4
WE-TVS-HS_SC70-6L_82400274
Center taps
VIRGINIA
INDUSTRY SOLUTIONS
Title cb-vis-acdc-v1.0
Size: A3
Number:9.
Revision:V1.0
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File: ETH Interface.SchDoc
Virginia industry solutions
11166 Fairfax Blvd
Fairfax
USA, VA
22030





VIRGINIA INDUSTRY SOLUTIONS

Title	cb-vis-acdc-v1.0		Virginia industry solutions 11166 Fairfax Blvd Fairfax USA, VA 22030
Size: A3	Number: 16.	Revision: V1.1	
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VIRGINIA INDUSTRY SOLUTIONS

Title	*cb-vis-acdc-v1.0*		*Virginia industry solutions* *11166 Fairfax Blvd* *Fairfax* *USA, VA* *22030*
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File: Debug.SchDoc			





IEC contactor, TeSys Deca, nonreversing, 32A resistive, 4 pole, 4 NO, 240VAC 50/60Hz coil, open style LC1DT32U7

TO POWER INPUT

J2aA — CON-1751529 — L1_IN, L2_IN, L3_IN, N_IN

F1a 2A, F2a 2A — PS1a IRM-10-12 — +V_PWR_IN — C12a 10uF 50V — GND_IN — GND

J2aB — CON-1751529 — L1_RELAY, L2_RELAY, L3_RELAY, N_RELAY

U2a 791002004: AC1_IN, AC1_OUT, AC2_IN, AC2_OUT, AC3_IN, AC3_OUT, AC4_IN, AC4_OUT, VCC, GND, NC, TEST_IN, ERROR_OUT, X6_OUT, X30_OUT, PWM_OUT

C14a 100nF 50V

+V5_SW — R5a 10K — RCD.TEST

+V3.3_SW — R7a 10K — RCD.ERROR

+V3.3_SW — R8a 10K — EMGC_SHDN_IN_3V3

+V3.3_SW — R55a 10K — T11aB NTZD3154NT1G — R56a 1K

R11a 10K — RCD.PWM

R12a 10K

R13a 10K — GND

Note: Würth part has 4.7V output on PWM, Benvac only 3.3V. For Benvac, only use 10k series resistor.

U1aA — IC-74LVC1G06 — Note: Benvac needs 5V on test input

U3aA — IC-SN74LVC1G02DCKR — EMGC_SHDN_IN

GPIO[0..3] — CCU_EMGC_SHDN_CMD, CHBG_1_GPIO_0, EMGC_SHDN_IN

If no system fault is detected, the output PIN 1 is at low level (GND). If a system fault is detected, PIN is at high impedance state. In this case, PINs 3 and 4 will be set to a high impedance state too (see tab.1).

If the residual current is below 6mA dc and no system fault occurs the output on PIN 3 is a low level (GND).

A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

SERIAL[0..3]

X20G — CON-Pionix-SOM — DownstreamComms: I3C.SDA, I3C.SCL, RS485_2_RX, RS485_2_TX, RS485_2_TXEN, CAN.HI, CAN.LO, RS485.A, RS485.B

CHBG_1_I2C-SDA, CHBG_1_I2C-SCL, CAN_HI, CAN_LO, RS485_P, RS485_N

R1a 470R — R2a 470R — D1a — D2a — MM3Z3V3ST1G

R3a 120R — R4a 470R — R6a 120R — R10a 470R

D3a MM3Z3V3ST1G

J5a — CON-Header-1x8-Male-1.5mm — GND, UART_RX, UART_TX, CAN_LO, CAN_HI, EMGC_SHDN_IN_3V3

U_ADCs — ADCs.SchDoc — ADC_TEMP[0..3]

X20A — CON-Pionix-SOM — Charging Cable: CONNECTOR.PIN0 … CONNECTOR.PIN5, CONNECTOR.RSVD

ADC_TEMP[0..3] — PT1000_2-, PT1000_2+, PT1000_1-, PT1000_1+, CP, PP, ZC

J3a — CON-Header-1x12-Male-1.5mm — +V_PWR_IN

X20I — CON-Pionix-SOM — USER INTERFACE: STOP_CHARGING, EMERGENCY_SHDN_IN, EMERGENCY_SHDN_OUT, LED.R, LED.G, LED.B

+V_SUPPLY_FILT — R16a 100K 1% — R18a 10K — STOP_CHARGING

D6a MM3Z3V3ST1G

Emergency shutdown Input. HI: Normal operation. LO: Trigger emergency shutdown. See section "Emergency input and output" for further details. Has internal 100kOhm pull down resistor, needs external pull up.

EMGC_SHDN_IN, EMGC_SHDN_OUT, LED_R, LED_G, LED_B

U6aA, U7aA, U7aB — IC-74LVC2G17 — R24a 120R, R25a 120R, R27a 22R

D9a — LED-150505M173300 — +V3.3_SW

+V_SUPPLY_FILT — C13a 100pF 50V

X20C — CON-Pionix-SOM — USER GPIO4 has a PWM timer — USER_GPIO0 … USER_GPIO8

RCD.TEST, RCD.ERROR, MOTOR_IN1, MOTOR_IN2, RCD.PWM, CHBG_1_GPIO_0, CHBG_1_GPIO_1, CHBG_1_GPIO_2

U4a DRV8871DDAR — IN1, IN2, VM, OUT1, OUT2, ILIM, GND — EMGC_SHDN_OUT

R14a 32.4kΩ — X1a — Male 1.5mm — MOTOR1, MOTOR2

R17a 10K — D5a MM3Z3V3ST1G

R19a 10K — D7a MM3Z3V3ST1G

Auxiliary contact required in the relays to IEC60947-4-1 mirror contact requirement

U1aB — IC-74LVC1G06 — +V3.3_SW — C15a 100nF 50V

J10a — CON-1751558 — Relay_Out1, Relay_Out2, Relay_Out3, Replay_FB_IN1, Replay_FB_IN2, Replay_FB_IN3

R26a 330R — R28a 330R — R29a 330R

U_Output interface1: Relay_Out1, Replay_FB_IN1, CHRG1, RELAY_FB1

U_Output interface2: Relay_Out2, Replay_FB_IN2, CHRG2, RELAY_FB2

U_Output interface3: Relay_Out3, Replay_FB_IN3, CHRG3, RELAY_FB3

X20H — CON-Pionix-SOM — Relays control: RELAY1, RELAY1_FB, RELAY2_FB, RELAY2, RELAY3, RELAY3_FB

U3aB — IC-SN74LVC1G02DCKR — +V3.3_SW — C16a 100nF 50V

U6aB — IC-74LVC2G17

U6aC — IC-74LVC2G17 — +V3.3_SW — C18a 100nF 50V

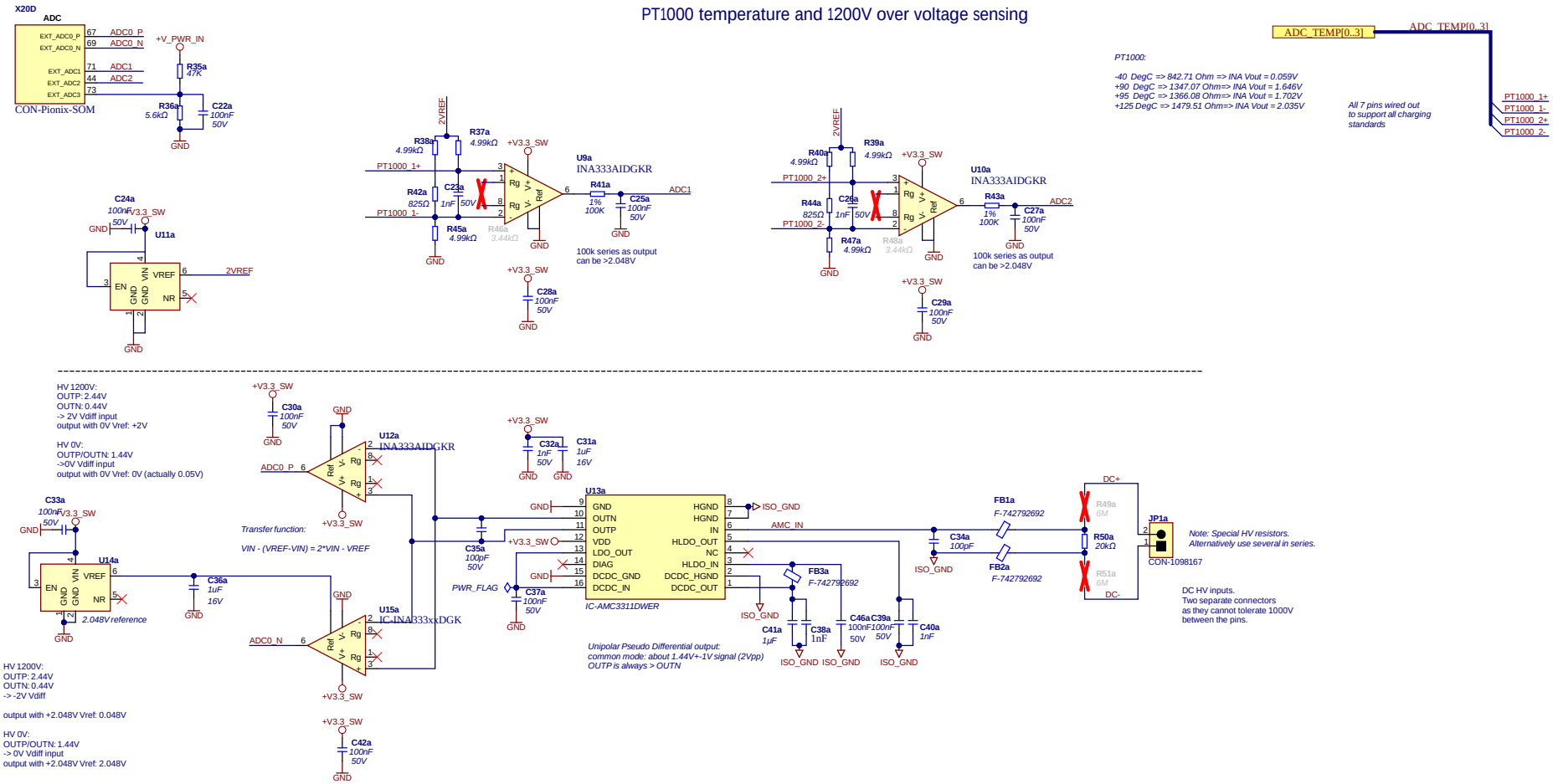
U7aC — IC-74LVC2G17 — +V3.3_SW — C19a 100nF 50V

X20B — CON-Pionix-SOM — +V3.3_SW — C21a 10uF 50V — GND, VDD3v3, SH1, SH2

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PT1000 temperature and 1200V over voltage sensing



HV 1200V:
OUTP: 2.44V
OUTN: 0.44V
-> 2V Vdiff input
output with 0V Vref: +2V

HV 0V:
OUTP/OUTN: 1.44V
->0V Vdiff input
output with 0V Vref: 0V (actually 0.05V)

HV 1200V:
OUTP: 2.44V
OUTN: 0.44V
-> -2V Vdiff

output with +2.048V Vref: 0.048V

HV 0V:
OUTP/OUTN: 1.44V
-> 0V Vdiff input
output with +2.048V Vref: 2.048V

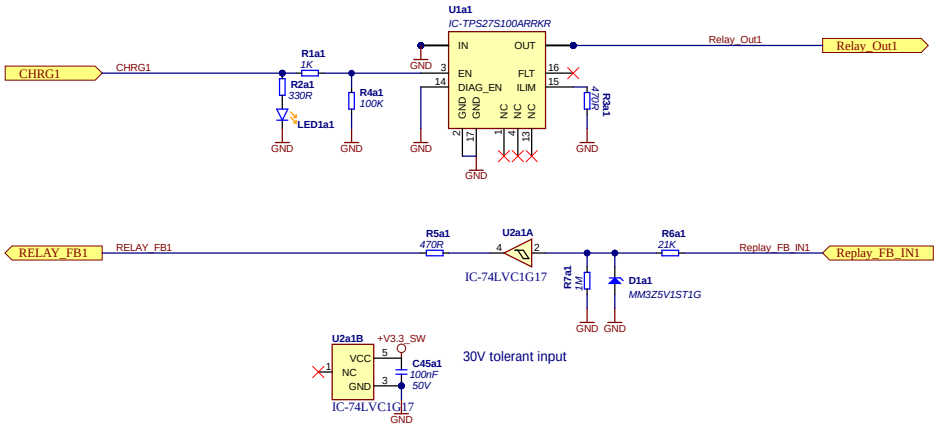
STM32 ADC input is 0 .. 2.048V on each pin (pseudo differential ADC)
AMC output is 1.44V +-1V on each channel for normal range,
max differential voltage for fail safe is -2.6V (will be out of range)

ISO15118-23:2023 requires 550V .. 1100V as OV detection limits.

During cert, a voltage of limit+15V will be applied to test the trigger point, but shutdown can occur slightly below limit voltage to account for errors as the shutdown limit is 10% above the max voltage limit.

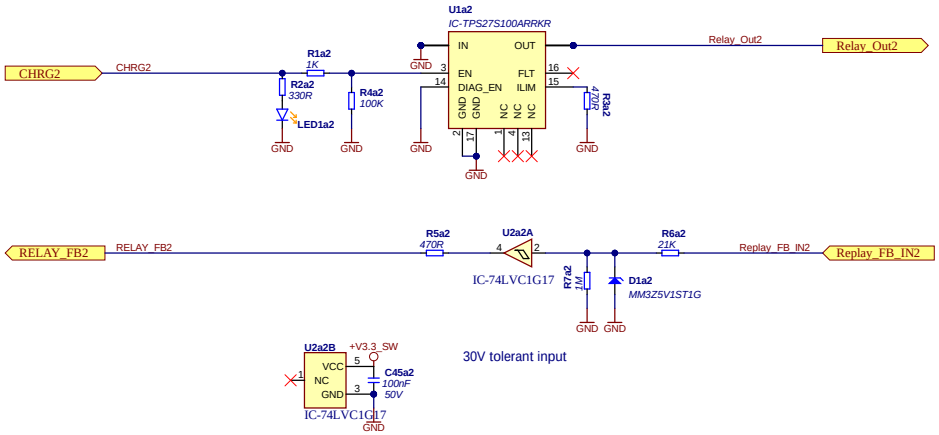
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Title: cb-vis-acdc-v1.0			Virginia industry solutions 11166 Fairfax Blvd Fairfax USA, VA 22030
Size: A3	Number: 17.	Revision: V1.0	
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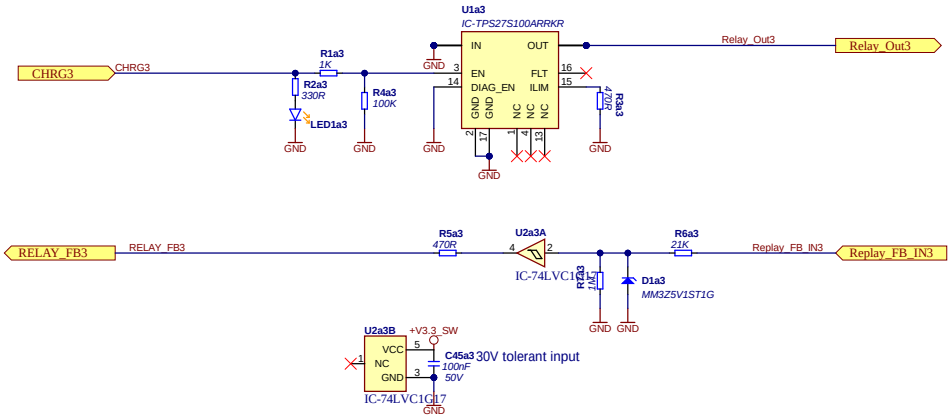


30V tolerant input

VIRGINIA INDUSTRY SOLUTIONS			*Virginia industry solutions* *11166 Fairfax Blvd* *Fairfax* *USA, VA* *22030*
Title **cb-vis-acdc-v1.0**			
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File: Output interface1.SchDoc			



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File: Output interface2.SchDoc			



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File: Output interface3.SchDoc			

HEATSINK STANDOFFS

SOM STANDOFFS



LVDS POWER JUMPER



PCB MOUNTING HOLES



VIRGINIA INDUSTRY SOLUTIONS

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